AI in Fiction vs Reality

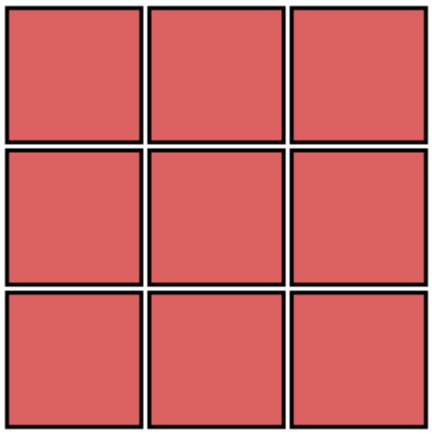
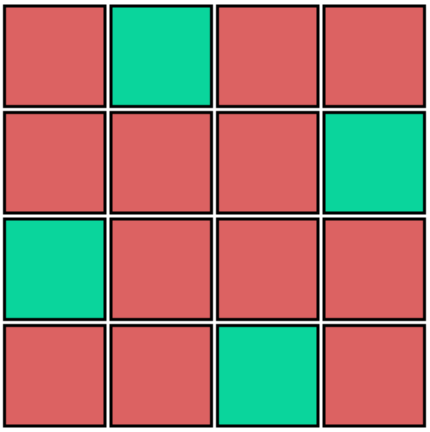
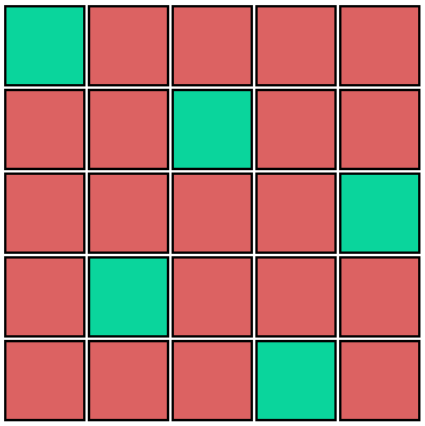
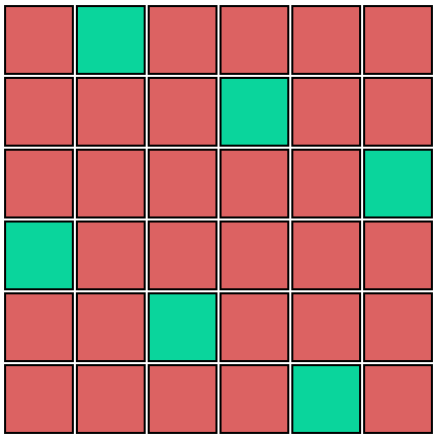
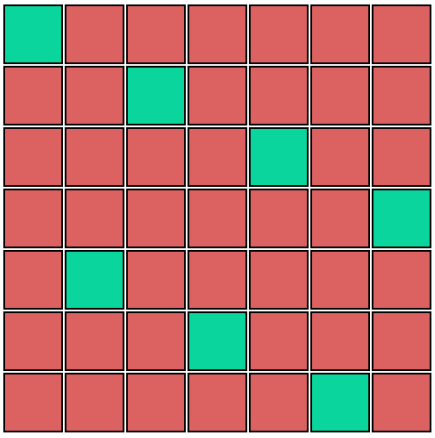
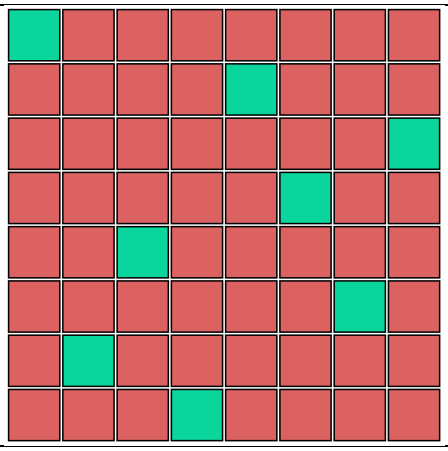
S. No.	Fiction	Reality
1	AI business automation will take over all our jobs.	AI aims to take over tasks, not whole jobs, especially the boring and repetitive ones.
2	AI might make societal inequalities and injustices worse.	AI systems perform based on the data we give them. If AI seems biased, it's reflecting the biases in the data.
3	AI can become self-aware.	AI systems work by following algorithms to process information and make decisions based on data.
4	AI can solve any problem, big or small.	AI depends on the data it uses, what specific problem it's trying to solve, and if it's using the right methods.
5	AI can make decisions on its own.	AI helps us make better decisions by sorting through a lot of data and offering insights we might miss.
6	AI will replace doctors completely.	AI assists doctors by analysing scans, predicting risks, and managing data.
7	AI can perfectly predict the future.	AI can only forecast trends using past data—it can't foresee unexpected events.
8	AI is 100% objective and neutral.	AI reflects the biases in the data it's trained on and the choices of its developers.
9	AI is always right.	AI can make mistakes, especially if trained on poor-quality or incomplete data.
10	AI doesn't need humans once it's built.	AI systems require ongoing monitoring, updates, and human oversight.
11	AI understands language like humans do.	AI processes patterns in text and speech but doesn't "understand" meaning in a human sense.
12	AI is a magical black box that just works.	AI is based on math, logic, and statistics—it's advanced but not magic.
13	AI has unlimited knowledge.	AI only knows what it's trained on—it doesn't "know everything."
14	AI can feel emotions like joy, anger, or sadness.	AI can simulate emotional responses but doesn't actually feel them.
15	AI learns exactly like humans.	AI learns from data patterns, not experiences, instincts, or context like humans.
16	AI can replace human creativity.	AI can generate art, music, or stories, but it lacks the depth of lived experience and intent behind human creativity.
17	AI will take control of governments and rule societies.	AI is a tool—its influence depends on how humans use it in decision-making.
18	AI can operate without any energy or resources.	AI requires massive computing power, storage, and electricity.
19	AI is immune to errors.	AI can fail if the input data is flawed or if the algorithm isn't suitable.
20	AI can replace teachers in schools.	AI supports teachers with personalized learning tools, but human mentorship and guidance are irreplaceable.
21	AI automatically improves over time without help.	AI models degrade if not retrained on new, updated data.
22	AI is only for tech companies.	AI is used across industries—healthcare, agriculture, education, transport, finance, and more.
23	AI always respects privacy.	AI can expose sensitive data if not carefully designed and regulated.
24	AI can instantly solve complex global crises.	AI is one tool among many—problems like climate change need political, social, and economic action too.
25	AI can understand human morality.	AI has no moral compass; it follows programmed rules and objectives.

History of AI

Groundwork for AI	
1921	Czech playwright Karel Čapek released a science fiction play "Rossum's Universal Robots" which introduced the idea of "artificial people" which he named robots. This was the first known use of the word.
1929	Japanese professor Makoto Nishimura built the first Japanese robot, named Gakutensoku.
1949	Computer scientist Edmund Callis Berkley published the book "Giant Brains, or Machines that Think" which compared the newer models of computers to human brains.
Birth of AI	
1950	Alan Turing published "Computer Machinery and Intelligence" which proposed a test of machine intelligence called The Imitation Game
1952	A computer scientist named Arthur Samuel developed a program to play checkers, which is the first to ever learn the game independently.
1955	John McCarthy held a workshop at Dartmouth on "artificial intelligence" which is the first use of the word, and how it came into popular usage.
AI Maturation	
1958	John McCarthy created LISP (acronym for List Processing), the first programming language for AI research, which is still in popular use to this day.
1959	Arthur Samuel created the term "machine learning" when doing a speech about teaching machines to play chess better than the humans who programmed them.
1961	The first industrial robot Unimate started working on an assembly line at General Motors in New Jersey, tasked with transporting die casings and welding parts on cars (which was deemed too dangerous for humans).
1965	Edward Feigenbaum and Joshua Lederberg created the first "expert system" which was a form of AI programmed to replicate the thinking and decision-making abilities of human experts.
1966	Joseph Weizenbaum created the first "chatterbot" (later shortened to chatbot), ELIZA, a mock psychotherapist, that used natural language processing (NLP) to converse with humans.1968: Soviet mathematician Alexey Ivakhnenko published "Group Method of Data Handling" in the journal "Avtomatika," which proposed a new approach to AI that would later become what we now know as "Deep Learning."
1973	An applied mathematician named James Lighthill gave a report to the British Science Council, underlining that strides were not as impressive as those that had been promised by scientists, which led to much-reduced support and funding for AI research from the British government.
1979	James L. Adams created The Standford Cart in 1961, which became one of the first examples of an autonomous vehicle. In '79, it successfully navigated a room full of chairs without human interference.
	The American Association of Artificial Intelligence which is now known as the Association for the Advancement of Artificial Intelligence (AAAI) was founded.
AI Boom	
1980	First conference of the AAAI was held at Stanford.
	The first expert system came into the commercial market, known as XCON (expert configurer). It was designed to assist in the ordering of computer systems by automatically picking components based on the customer's needs.
1981	The Japanese government allocated $850 million (over $2 billion dollars in today's money) to the Fifth Generation Computer project. Their aim was to create computers that could translate, converse in human language, and express reasoning on a human level.
1984	The AAAI warns of an incoming "AI Winter" where funding and interest would decrease, and make research significantly more difficult.
1985	An autonomous drawing program known as AARON is demonstrated at the AAAI conference.
1986	Ernst Dickmann and his team at Bundeswehr University of Munich created and demonstrated the first driverless car (or robot car). It could drive up to 55 mph on roads that didn't have other obstacles or human drivers.
1987	Commercial launch of Alacrity by Alactrious Inc. Alacrity was the first strategy managerial advisory system, and used a complex expert system with 3,000+ rules.
AI Winter	
1987	The market for specialized LISP-based hardware collapsed due to cheaper and more accessible competitors that could run LISP software, including those offered by IBM and Apple. This caused many specialized LISP companies to fail as the technology was now easily accessible.
1988	A computer programmer named Rollo Carpenter invented the chatbot Jabberwacky, which he programmed to provide interesting and entertaining conversation to humans.
AI Agents	
1997	Deep Blue (developed by IBM) beat the world chess champion, Gary Kasparov, in a highly-publicized match, becoming the first program to beat a human chess champion.
	Windows released a speech recognition software (developed by Dragon Systems).
2000	Professor Cynthia Breazeal developed the first robot that could simulate human emotions with its face,which included eyes, eyebrows, ears, and a mouth. It was called Kismet.
2002	The first Roomba was released.
2003	Nasa landed two rovers onto Mars (Spirit and Opportunity) and they navigated the surface of the planet without human intervention.
2006	Companies such as Twitter, Facebook, and Netflix started utilizing AI as a part of their advertising and user experience (UX) algorithms.
2010	Microsoft launched the Xbox 360 Kinect, the first gaming hardware designed to track body movement and translate it into gaming directions.
2011	An NLP computer programmed to answer questions named Watson (created by IBM) won Jeopardy against two former champions in a televised game.
	Apple released Siri, the first popular virtual assistant.
Artificial General Intelligence	
2012	Two researchers from Google (Jeff Dean and Andrew Ng) trained a neural network to recognize cats by showing it unlabeled images and no background information.
2015	Elon Musk, Stephen Hawking, and Steve Wozniak (and over 3,000 others) signed an open letter to the worlds' government systems banning the development of (and later, use of) autonomous weapons for purposes of war.
2016	Hanson Robotics created a humanoid robot named Sophia, who became known as the first "robot citizen" and was the first robot created with a realistic human appearance and the ability to see and replicate emotions, as well as to communicate.

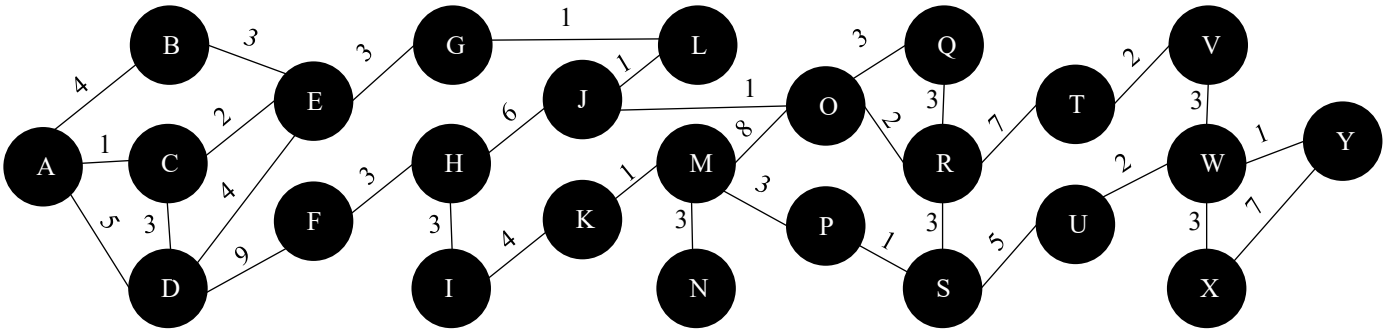
2017	Facebook programmed two AI chatbots to converse and learn how to negotiate, but as they went back and forth they ended up forgoing English and developing their own language, completely autonomously.
2018	A Chinese tech group called Alibaba's language-processing AI beat human intellect on a Stanford reading and comprehension test.
2019	Google's AlphaStar reached Grandmaster on the video game StarCraft 2, outperforming all but .2% of human players.
2020	OpenAI started beta testing GPT-3, a model that uses Deep Learning to create code, poetry, and other such language and writing tasks. While not the first of its kind, it is the first that creates content almost indistinguishable from those created by humans.
2021	OpenAI developed DALL-E, which can process and understand images enough to produce accurate captions, moving AI one step closer to understanding the visual world.
Recent Developments	
2022	MidJourney launches, an AI art generator that creates digital images from text prompts. It quickly becomes popular among artists, designers, and hobbyists for its creative flexibility.
	Stable Diffusion is released as an open-source text-to-image model. Its accessibility sparks a massive wave of experimentation and debate around AI art.
	OpenAI launches ChatGPT, a conversational AI based on GPT-3.5. It becomes the first LLM widely adopted by the public, reaching millions of users in weeks.
2023	GPT-4 is released, offering major improvements in reasoning, creativity, and context length. It shows remarkable performance on academic benchmarks and coding tasks.
	OpenAI launches the ChatGPT API. This allows developers to integrate conversational AI into apps and services, accelerating adoption across industries.
	Anthropic launches Claude, a rival AI assistant designed with safety in mind. It is named after Claude Shannon, the father of information theory.
	Google introduces Bard, later rebranded as Gemini. It marks Google's entry into generative AI chatbots and becomes integrated into Google's ecosystem.
	Adobe debuts Firefly, a generative AI tool for images and design. It emphasizes ethical training data and is embedded into Adobe Creative Cloud.
	Meta releases LLaMA, an open-source large language model. It quickly becomes a foundation for many community-driven AI projects.
	U.S. President Joe Biden signs an AI Executive Order. It emphasizes innovation, national security, and protections against AI bias.
	The UK hosts the AI Safety Summit at Bletchley Park. Governments, researchers, and companies discuss risks and coordinate global safety standards.
2024	Stability AI releases Stable LM 2, a next-generation open-source language model. It improves reasoning and efficiency compared to earlier releases.
	OpenAI unveils Sora, a powerful text-to-video model. It can generate realistic video clips from written prompts, pushing AI into dynamic media.
	Google launches Gemini 1.5 with a context window of up to 1 million tokens. This enables AI to process entire books, codebases, or long conversations.
	DeepMind releases AlphaFold 3, expanding beyond protein folding to model a wide range of biomolecules. It is considered a major breakthrough for biology and medicine.
	The EU AI Act officially comes into force. It establishes the world's first comprehensive legal framework for AI use.
2025	DeepSeek-R1 is released as a powerful open-source LLM. It demonstrates the competitiveness of AI research outside the U.S. and China.
	Microsoft, OpenAI, and NVIDIA announce the Stargate Project. It is a multi-billion-dollar supercomputer effort to train future frontier AI models.
	Researchers publish "Humanity's Last Exam," a benchmark for human-level reasoning in AI. It becomes a new standard for testing AGI progress.
	OpenAI introduces Deep Research for ChatGPT. This tool allows AI to perform extended online research autonomously.
	Mistral AI launches "Le Chat," a European alternative to ChatGPT. It highlights Europe's growing influence in the AI race.
	World leaders gather at the Paris AI Action Summit. They announce the €200B InvestAI initiative to accelerate AI research and adoption.
	OpenAI previews GPT-4.5. It offers faster reasoning and efficiency, serving as a bridge toward GPT-5.
	OpenAI launches o3 and o4-mini models. These are lightweight, specialized models optimized for cost and speed.
	Google unveils AlphaEvolve, Veo 3, and Gemini 2.5 Pro. These models advance scientific discovery, video generation, and reasoning abilities.
	Anthropic releases Claude 4. It offers improved reliability, long context, and safer reasoning.
	OpenAI releases GPT-5. It introduces major improvements in reasoning, planning, and tool use, though debates arise over scaling limits.

AI Bot – N-Queens
N = 3 to N = 8

3		4	
5		6	
7		8	

Uninformed Search

Initial State – A
Goal State - Y



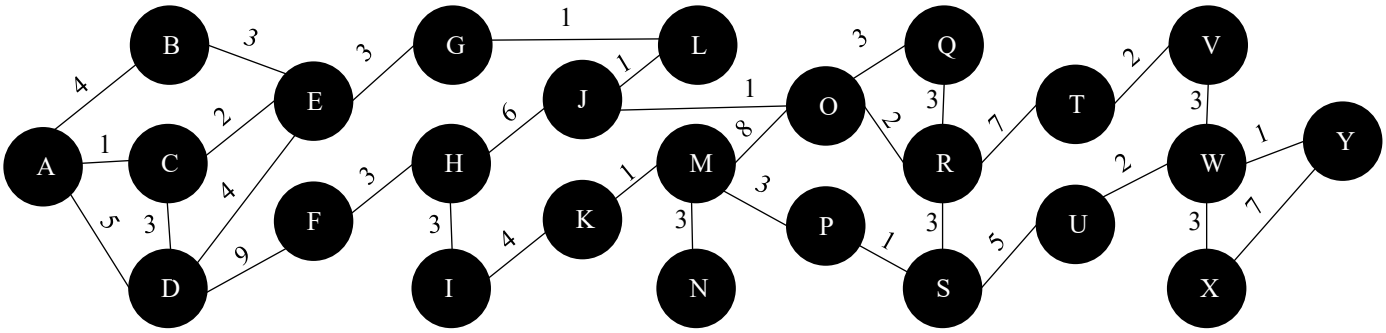
1) Breadth First Search

Iteration	Traversed Path	Frontier – Queue
1		A
2	A	B, C, D
3	A, B	C, D, E
4	A, B, C	D, E
5	A, B, C, D	E, F
6	A, B, C, D, E	F, G
7	A, B, C, D, E, F	G, H
8	A, B, C, D, E, F, G	H, L
9	A, B, C, D, E, F, G, H	L, I, J
10	A, B, C, D, E, F, G, H, L	I, J
11	A, B, C, D, E, F, G, H, L, I	J, K
12	A, B, C, D, E, F, G, H, L, I, J	K, O
13	A, B, C, D, E, F, G, H, L, I, J, K	O, M
14	A, B, C, D, E, F, G, H, L, I, J, K, O	M, Q, R
15	A, B, C, D, E, F, G, H, L, I, J, K, O, M	Q, R, N, P
16	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q	R, N, P
17	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R	N, P, S, T
18	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N	P, S, T
19	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P	S, T
20	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S	T, U
21	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T	U, V
22	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T, U	V, W
23	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T, U, V	W, X, Y
24	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T, U, V, W	X, Y
25	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T, U, V, W, X	Y
26	A, B, C, D, E, F, G, H, L, I, J, K, O, M, Q, R, N, P, S, T, U, V, W, X, Y	

Goal state reached in iteration 26.

Uninformed Search

Initial State – A
Goal State - Y



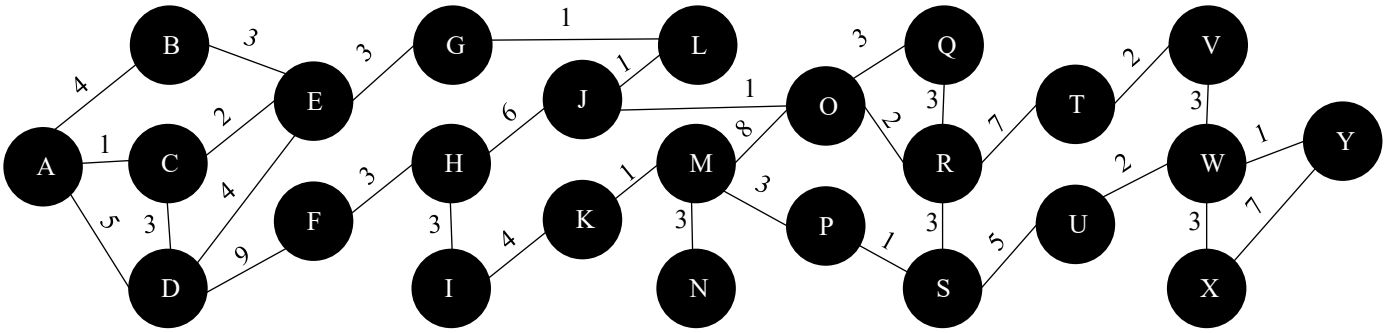
2) Uniform Cost Search

Iteration	Node	Visited	Cost	Frontier – Priority Queue
1				A=0
2	A	A	0	B^{A}=4, C^{A}=1, D^{A}=5
3	C^{A}	A, C	1	B^{A}=4, D^{A}=5, D^{CA}=1+3 E^{CA}=1+2
4	E^{CA}	A, C, E	3	B^{A}=4, B^{ECA}=3+3, D^{CA}=4, D^{ECA}=3+4, G^{ECA}=3+3
5	B^{A}	A, C, E, B	4	D^{CA}=4, G^{ECA}=6
6	D^{CA}	A, C, E, B, D	4	F^{DCA}=4+9, G^{ECA}=6
7	G^{ECA}	A, C, E, B, D	6	F^{DCA}=13, L^{GECA}=6+1
8	L^{GECA}	A, C, E, B, D, L	7	F^{DCA}=13, J^{LGECA} =7+1
9	J^{LGECA}	A, C, E, B, D, L, J	8	F^{DCA}=13, H^{JLGECA}=8+6, O^{JLGECA}=8+1
10	O^{JLGECA}	A, C, E, B, D, L, J, O	9	F^{DCA}=13, H^{JLGECA}=14, $M^{OJLGECA}$=9+8, Q^{OLGECA}=9+3, $R^{OJLGECA}$=9+2
11	$R^{OJLGECA}$	A, C, E, B, D, L, J, O, R	11	F^{DCA}=13, H^{JLGECA}=14, $M^{OJLGECA}$=17, Q^{OLGECA}=12, $S^{ROJLGECA}$=11+3, $T^{ROJLGECA}$=11+7
12	Q^{OLGECA}	A, C, E, B, D, L, J, O, R, Q	12	F^{DCA}=13, H^{JLGECA}=14, $M^{OJLGECA}$=17, $S^{ROJLGECA}$=14, $T^{ROJLGECA}$=18
13	F^{DCA}	A, C, E, B, D, L, J, O, R, Q, F	13	H^{JLGECA}=14, H^{FDCA}=13+3, $M^{OJLGECA}$=17, $S^{ROJLGECA}$=14, $T^{ROJLGECA}$=18
14	H^{JLGECA}	A, C, E, B, D, L, J, O, R, Q, F, H	14	$I^{HJLGECA}$=14+3, $M^{OJLGECA}$=17, $S^{ROJLGECA}$=14, $T^{ROJLGECA}$=18
15	$S^{ROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S	14	$I^{HJLGECA}$=17, $M^{OJLGECA}$=17, $P^{SROJLGECA}$=14+1, $T^{ROJLGECA}$=18, $U^{SROJLGECA}$=14+5
16	$P^{SROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P	15	$I^{HJLGECA}$=17, $M^{OJLGECA}$=17, $T^{ROJLGECA}$=18, $U^{SROJLGECA}$=14+5
17	$I^{HJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I	17	$K^{IHJLGECA}$=17+4, $M^{OJLGECA}$=17, $T^{ROJLGECA}$=18, $U^{SROJLGECA}$=19
18	$M^{OJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M	17	$N^{MOJLGECA}$=17+3, $T^{ROJLGECA}$=18, $U^{SROJLGECA}$=19
19	$K^{MOJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K	18	$N^{MOJLGECA}$=20, $T^{ROJLGECA}$=18, $U^{SROJLGECA}$=19
20	$T^{ROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T	18	$N^{MOJLGECA}$=20, $U^{SROJLGECA}$=19, $V^{TROJLGECA}$=18+2
21	$U^{SROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T, U	19	$N^{MOJLGECA}$=20, $V^{TROJLGECA}$=20, $W^{USROJLGECA}$=19+2
22	$N^{MOJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T, U, N	20	$V^{TROJLGECA}$=20, $W^{USROJLGECA}$=21
23	$V^{TROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T, U, N, V	20	$W^{USROJLGECA}$=21, $W^{VTROJLGECA}$=20+3
24	$W^{USROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T, U, N, V, W	21	$X^{WUSROJLGECA}$=21+3, $Y^{WUSROJLGECA}$=21+1
25	$Y^{WUSROJLGECA}$	A, C, E, B, D, L, J, O, R, Q, F, H, S, P, I, M, K, T, U, N, V, W, Y	22	$X^{WUSROJLGECA}$=24

Goal state reached.
Shortest Path: A→C→E→G→L→J→O→R→S→U→W→Y
Cost: 22

Uninformed Search

Initial State – A
Goal State - Y



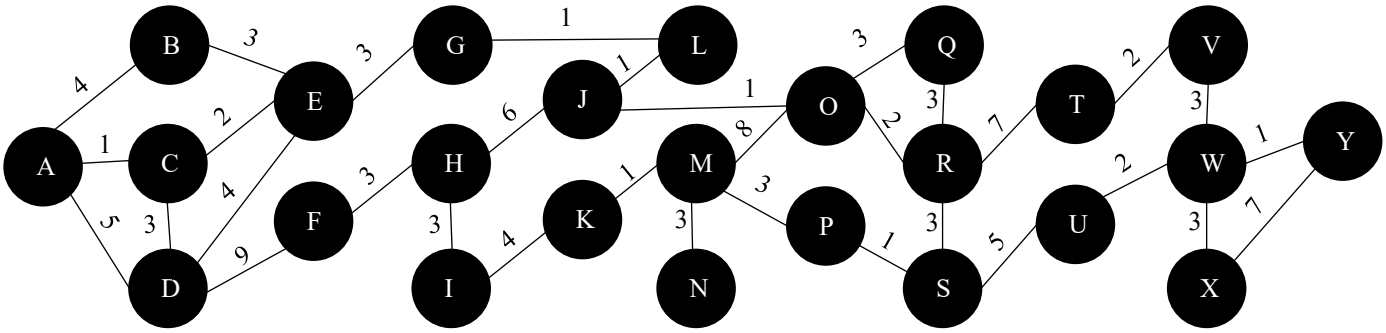
3) Depth First Search

Iteration	Traversed Path	Frontier – Stack
1		A
2	A	D, C, B
3	A, B	D, C, E
4	A, B, E	D, C, G
5	A, B, E, G	D, C, L
6	A, B, E, G, L	D, C, J
7	A, B, E, G, L, J	D, C, O, H
8	A, B, E, G, L, J, H	D, C, O, I, F
9	A, B, E, G, L, J, H, F	D, C, O, I
10	A, B, E, G, L, J, H, F, I	D, C, O, K
11	A, B, E, G, L, J, H, F, I, K	D, C, O, M
12	A, B, E, G, L, J, H, F, I, K, M	D, C, O, P, N
13	A, B, E, G, L, J, H, F, I, K, M, N	D, C, O, P
14	A, B, E, G, L, J, H, F, I, K, M, N, P	D, C, O, S
15	A, B, E, G, L, J, H, F, I, K, M, N, P, S	D, C, O, U, R
16	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R	D, C, O, U, T, Q
17	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q	D, C, O, U, T
18	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T	D, C, O, U, V
19	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V	D, C, O, U, W
20	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W	D, C, O, U, Y, X
21	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X	D, C, O, U, Y
22	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X, Y	D, C, O, U
23	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X, Y, U	D, C, O
24	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X, Y, U, O	D, C
25	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X, Y, U, O, C	D
26	A, B, E, G, L, J, H, F, I, K, M, N, P, S, R, Q, T, V, W, X, Y, U, O, C, D	

Goal state reached in iteration 22.

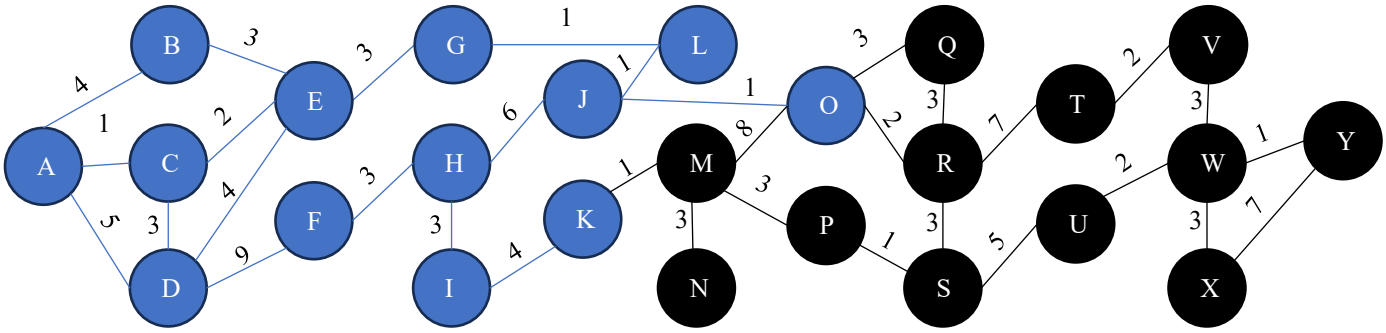
Uninformed Search

Initial State – A
Goal State - Y



4) Depth Limited Search

Limit = 4

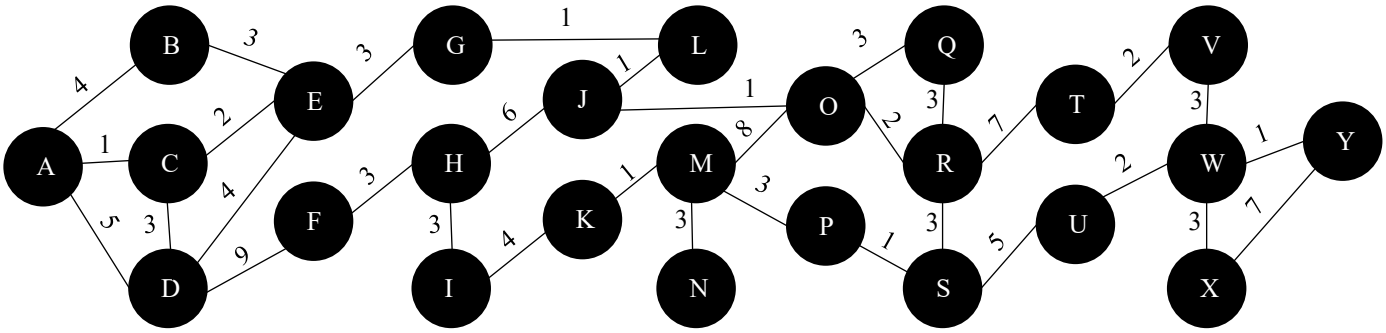


Iteration	Traversed Path	Frontier – Stack
1		A
2	A	D, C, B
3	A, B	D, C, E
4	A, B, E	D, C, G
5	A, B, E, G	D, C, L
6	A, B, E, G, L	D, C, J
7	A, B, E, G, L, J	D, C
8	A, B, E, G, L, J, C	D
9	A, B, E, G, L, J, C, D	F
10	A, B, E, G, L, J, C, D, F	H
11	A, B, E, G, L, J, C, D, F, H	J, I
12	A, B, E, G, L, J, C, D, F, H, I	J, K
13	A, B, E, G, L, J, C, D, F, H, I, K	J
14	A, B, E, G, L, J, C, D, F, H, I, K	O
15	A, B, E, G, L, J, C, D, F, H, I, K, O	

Goal state NOT reached.

Uninformed Search

Initial State – A
Goal State - Y



5) Iterative Deepening Search

Limit = 0

Iteration	Traversed Path	Frontier – Stack
1		A
2	A	

Goal state NOT reached.

Limit = 1

Iteration	Traversed Path	Frontier – Stack
1		A
2	A	D, C, B
3	A, B	D, C
4	A, B, C	D
5	A, B, C, D	

Goal state NOT reached.

Limit = 2

Iteration	Traversed Path	Frontier – Stack
1		A
2	A	D, C, B
3	A, B	D, C, E
4	A, B, E	D, C
5	A, B, E, C	D
6	A, B, E, C, D	F
7	A, B, E, C, D, F	

Goal state NOT reached.

Limit = 3

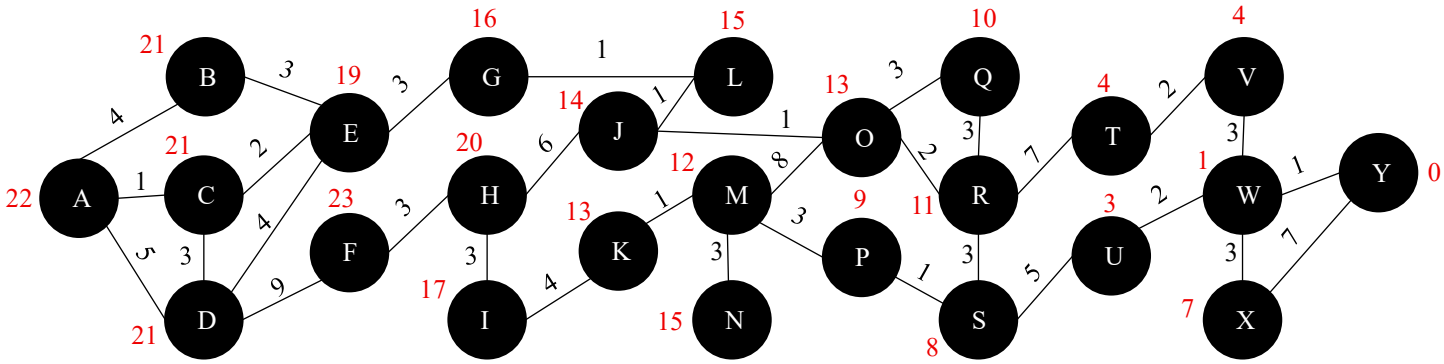
Iteration	Traversed Path	Frontier – Stack
1		A
2	A	D, C, B
3	A, B	D, C, E
4	A, B, E	D, C, G
5	A, B, E, G	D, C
6	A, B, E, G, C	D
7	A, B, E, G, C, D	F
8	A, B, E, G, C, D, F	G
9	A, B, E, G, C, D, F, G	

Goal state NOT reached.

Informed Search

Initial State – A
Goal State – Y

g(n) in BLACK
h(n) in RED



1) Greedy Best-First Search

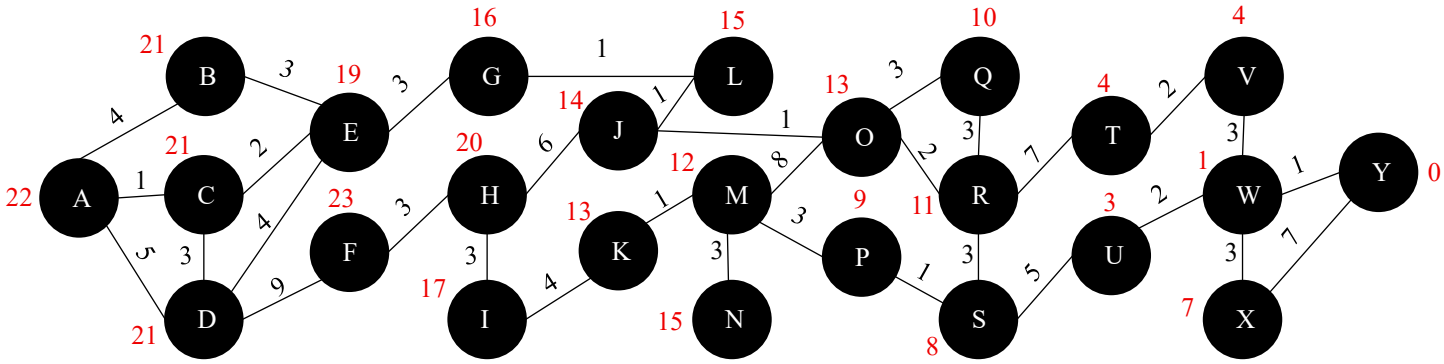
Iteration	Node	Cost	Frontier – Priority Queue wrt h(n)
1			A=22
2	A	0	B=21, C=21, D=21
3	B	0+4=4	E=19
4	E	4+3=7	D=21, G=16
5	G	7+3=10	L=15
6	L	10+1=11	J=14
7	J	11+1=12	H=20, O=13
8	O	12+1=13	M=12, Q=10, R=11
9	Q	13+3=16	R=11
10	R	16+3=19	S=8, T=4
11	T	19+7=26	V=4
12	V	26+2=28	W=1
13	W	29+3=32	X=7, Y=0
14	Y	32+1=33	

Goal state reached.
Shortest Path: A→B→E→G→L→J→O→Q→R→T→V→W→Y
Cost: 33

Informed Search

Initial State – A
Goal State – Y

g(n) in BLACK
h(n) in RED



2) A* Search

If frontier costs are same, the one with the smaller g(n) is prioritised.

Iteration	Node	Cost – g(n)	Frontier – Priority Queue wrt f(n) = g(n) + h(n)
1			A=0+22=22
2	A	0	B=4+21=25, C=1+21=22, D=5+21=26
3	C	1	E=(1+2)+19=22
4	E	3	D=(3+4)+21=28, G=(3+3)+16=22
5	G	6	L=(6+1)+15=22
6	L	7	J=(7+1)+14=22
7	J	8	H=(8+6)+20=34, O=(8+1)+13=22
8	O	9	M=(9+8)+12=29, Q=(9+3)+10=22, R=(9+2)+11=22
9	R	11	S=(11+3)+8=22, T=(11+7)+4=22
10	S	14	P=(14+9)+1=24, U=(14+5)+3=22
11	U	19	W=(19+2)+1=22
12	W	21	V=(21+3)+4=28, X=(21+3)+7=31, Y=(21+1)+0=22
13	Y	22	

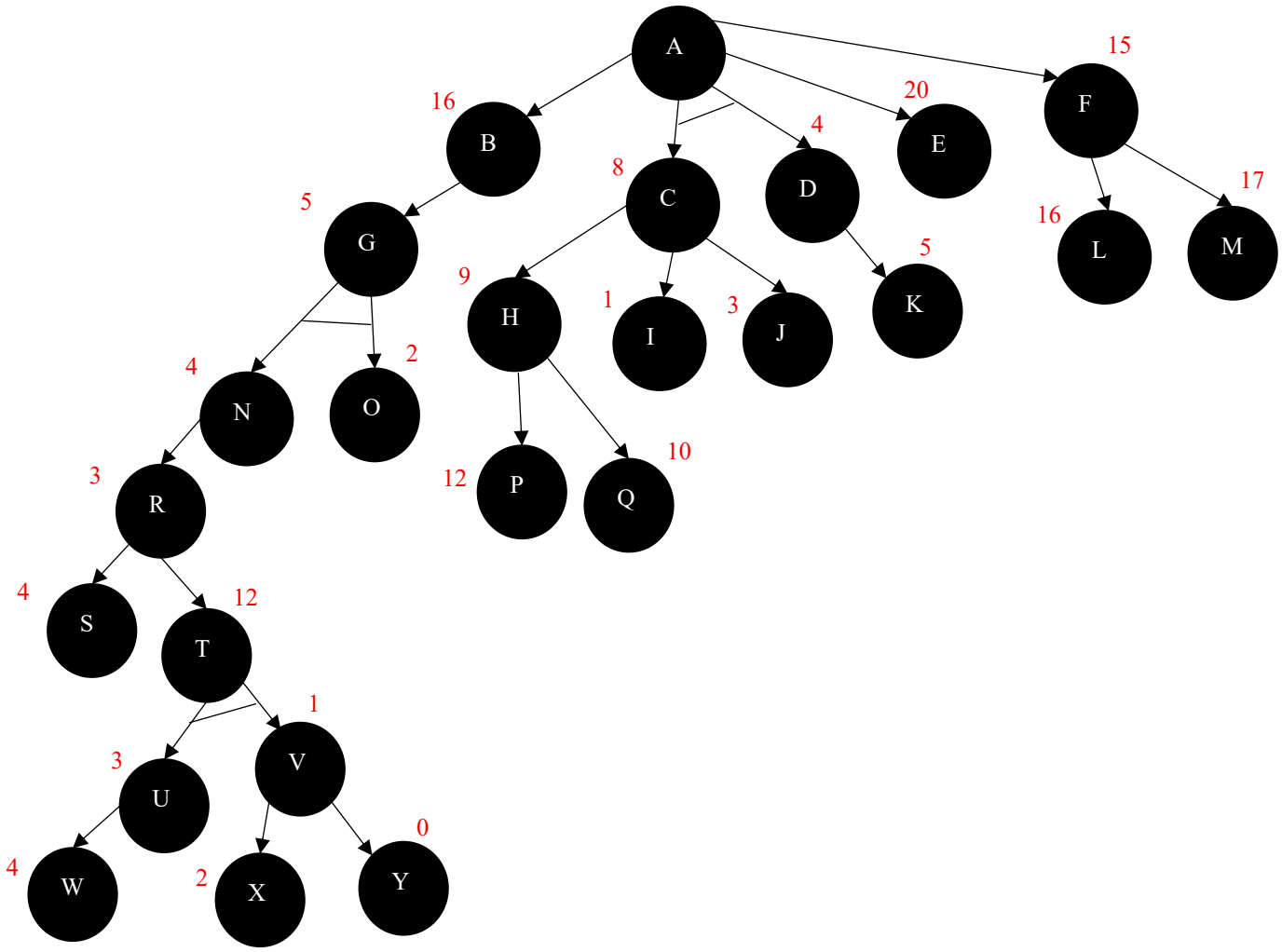
Goal state reached.
Shortest Path: A→C→E→G→L→J→O→R→S→U→W→Y
Cost: 22

Informed Search

Initial State – A
Goal State – Y

Edge cost is 1 for all edges
h(n) in RED

3) AO* Search



Summary of Different Search Paradigms

Type	Method	Goal State Found?	Iterations	Path	Cost	Optimal?
Uninformed Search	Breadth First Search	Yes	26	A→B→C→D→E→F→G→H→L→I→J→K→O→M →Q→R→N→P→S→T→U→V→W→X→Y	-	No
	Uniform Cost Search	Yes	25	A→C→E→G→L→J→O→R→S→U→W→Y	22	Yes
	Depth First Search	Yes	22	A→B→E→G→L→J→H→F→I→K→M→N→P→S →R→Q→T→V→W→X→Y		
	Depth Limited Search	No (for l = 4)	-	-		
	Iterative Deepening Search	No (l=0 to 3)	-	-		
Informed Search	Greedy Best-First Search	Yes	14	A→B→E→G→L→J→O→Q→R→T→V→W→Y	33	No
	A* Search	Yes	13	A→C→E→G→L→J→O→R→S→U→W→Y	22	Yes
	AO* Search (diff. graph)	Yes	-	A→B→G→N→R→T→V→Y	7	Yes